

Design and characterization of a pneumatic muscle actuator with novel end-fittings for medical assistive applications

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ABSTRACT

Pneumatic muscle actuators (PMA) are a class of soft actuators known for their high power to weight ratio and inherent compliance. The pneumatic muscle's inherent properties make them very favorable for assistive applications (e.g., medical exoskeletons). This study presents a novel end-fitting design that makes the developed pneumatic muscle actuator lightweight, cost-effective, and modular, thus simplifying the process of assembly and maintenance. The pneumatic muscle actuator assembled using the novel end fittings achieves a shorter overall length without compromising its contraction. The pneumatic muscle actuator has been assembled using a commercial bladder and a braided sleeve alongside a pair of 3D printed novel end-fittings. The paper also details the developed actuator's characterization for force and deflection parameters at various operating pressures. A total of four muscle actuators of varying diameters with constant actuation length (100 mm) were developed and tested to showcase the effect of size on the muscle actuator's behavior. The study presented here also involved comparing three mathematical models developed for pneumatic muscles in order to find a model which closely resembles the developed muscle actuator. Finally, the developed pneumatic muscle actuator's behavior is compared with a commercially available muscle to determine the efficacy of the developed muscle's design. The tests showed that the muscle using a bladder of smaller volume but higher tensile modulus had a higher accuracy and stable performance. As the muscle is intended for medical applications, it was also put through an endurance test with realistic loading and pressure conditions, which revealed very promising results.

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1. Introduction

The McKibben muscle a type of pneumatic muscle actuator (PMA), consists of three essential components: a hyper-elastic bladder, a braided sleeve enclosing the bladder, and two end fittings clamped on either end. The history of pneumatic muscles reveals two significant events, the first was its invention by Richard Gaylord in 1958 [1], but the inability of the muscle actuator to produce high force and the requirement of a power pack (compressor) led to it being side-lined by the electric motor [2]. Joseph L McKibben reintroduced the PMA in the year 1960; as a result, the PMA is also known as the McKibben Muscle. The principle of operation of the PMA involves the supply of pressurized air which results in the radial expansion of the bladder, which in turn is restricted by the enveloping braided sleeve, thus converting the radial motion

into axial contraction [3]. There are two output parameters linked with PMAs, which include maximum axial contraction and blocked force. Maximum axial contraction is realized by allowing the actuator to actuate freely with its ends unconstrained, while the force generated by an actuated muscle whose ends are constrained is known as blocked force. The PMA's high power to weight ratio [4] and inherent compliance make it very favorable for assistive applications such as medical exoskeletons. Daerden et al. [5] defined compliance as the inverse of stiffness, which is a result of the compressible nature of the fluid (air) filled within the bladder [6] and the hyperelastic nature of the bladder.

Several researchers have worked on developing a compact and efficient design of the PMA. Kothera et al. [7] aimed at presenting the experimental characterization of McKibben muscles of varying length and diameter and compared the results with a mathematical formulation of the muscle. The improved mathematical model developed was able to predict the static behavior of the muscle with greater accuracy. Hocking et al. [8] developed a miniature pneumatic muscle intending to use it in Robotics and Aerospace applications. They developed, fabricated,

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and experimentally characterized a miniature-sized pneumatic muscle actuator. An improved version of the force block model was developed and found to predict the behavior of the miniature PMA with high accuracy. A study on miniature PMAs was also conducted by De et al. [9], who developed a control strategy for the accurate positioning of their developed PMA and compared the results with a model simulation. Vocke et al. [10] designed and tested a small scale pneumatic muscle for micro-air vehicle (MAV) applications. They also presented a comparison between PMA and conventional electric servo actuators for MAV applications. Koizumi et al. [11] developed a soft actuator by braiding a number of McKibben muscles together. The study showed an increase in the contraction percentage of up to 37 %. The main drawback of the multi-braided actuator was the decrease in contraction force. Oliver et al. [12] developed and characterized a braided pneumatic muscle and used it to actuate a four degree of freedom mechatronic finger. The study involved the fabrication of three muscle lengths (60 mm, 100 mm, and 140 mm) with a constant diameter of 6 mm, which were tested up to a pressure of 4 bar and payload of 10Kg. The author reported a life of 260,000 cycles at a 14% contraction per cycle, after which perforation of the silicon bladder was observed. Chakravarthy et al. [13] developed a small braided pneumatic muscle having a diameter of 1.2 mm. The muscle was developed and experimentally characterized and used to actuate an endoscopic surgical tool. Wakimoto et al. [14] developed a flexible muscle actuator using a number of smaller diameter McKibben muscles in a bundle. The study also involved comparing the experimental results with those obtained through FEM analysis. Another research oriented on a combination of pneumatic muscles was conducted by Peng et al. [15] They developed and characterised a thin pneumatic muscle and used it to actuate a light weight and compliant robotic manipulator. Antonelli et al. [16] developed a finite element model of the McKibben muscle and validated its results with experimental values. They considered the non-linearity of the inner tube using Mooney Rivlin equation, though they neglected friction between the braid and tube they were able to match the experimental values with good accuracy. Wang et al. [17] derived a quasi-static model for the McKibben muscle actuator in order to predict the force and contraction accurately with respect to the Gaylord's formulation. The formulation was validated with experimental results of muscles of different lengths.

Literature reveals a focus on research involving pneumatic muscle with short actuation lengths, which is excellent for applications requiring micro-actuation. The contraction percentage produced by a McKibben muscle is independent of its length and is dependent only on the parameters of the braided sleeve (e.g. braid angle) and bladder (e.g. tensile modulus) used. As seen in the literature, the pneumatic muscles developed are seen to produce low levels of contraction percentage and blocked force. In larger applications such as assistive exoskeletons, the miniature muscle falls short in the amount of contraction and load handling capability. The behavior of the pneumatic muscle is highly influenced by the bladder and braided sleeve used in its construction. As the bladder occupies a large volume percentage of the PMA, its material, and geometric properties greatly influence the behavior of the PMA. The other drawback of the existing pneumatic muscles is the lack of flexibility and difficulty in assembly/disassembly, which significantly affects experimenting with new bladder and braided sleeve materials and maintenance operation. A number of researchers strived to ease the process of assembly of the PMA, but the use of metal fittings increases the weight of the overall muscle, and there is a risk of damage to the delicate bladder. This paper presents a novel end-fitting arrangement for easy assembly and maintenance of the PMA, allowing the user the choice of selecting the bladder and braided sleeve based on the intended application.

A mathematical model can optimize the process of finding the right size and material of the bladder and braided sleeve, which best suits the intended application (e.g., upper limb exoskeleton). The PMA being a soft actuator using hyperelastic bladder poses challenges in modelling mathematically. The literature reveals several researchers attempting to model the PMA using techniques such as energy or force balance [1,3,4,7,18–20]. These models were later improved by considering additional factors such as the effect of the braided sleeve (e.g., friction between braids) and bladder (e.g., Hyperelastic nature) on the force or contraction developed by the PMA [3,4,7,16,18–20]. This paper presents the comparison of three mathematical models in order to find a model which closely resembles the behavior of the proposed PMA.

The paper is divided into four sections. The second section presents the design and assembly of the developed PMA using the novel end-fittings and clamps. In the third section, the characterization of the developed PMA for force and contraction percentage with respect to actuation pressure is presented. The third section also compares the results of the developed PMA with a commercially available PMA. The fourth section presents an evaluation of various mathematical models to find the one which best represents the fabricated PMA and compare the experimental results with those obtained from the mathematical simulation. As the PMA is being designed for medical applications, the last section highlights and tests the muscle's efficacy through an endurance test, and results obtained showed a highly promising behavior of the developed PMA.

2. Design and fabrication

The basic structure of the PMA designed in this work consists of three main components, the flexible bladder, the braided sleeve, and the end fittings. The choice of the bladder and braided sleeve significantly influences the behavior (force and contraction) of the PMA. An application that requires to manipulate large loads (eg 5Kg payload to actuate elbow exoskeleton) requires a thicker bladder with a bigger diameter [21]. As the bladder's size and the braided sleeve's initial braid angle play an essential role in determining the maximum contraction percentage achieved by the PMA [3], it is beneficial to allow the user to select and change the same based on the requirements of the application. The ease of assembly and disassembly demands the PMA's design to be modular and which also makes it easy to maintain. In this work, the novel end-fittings and clamps which allow for easy assembly and disassembly of the muscle are designed and incorporated.

The PMA design proposed in the current work consists of two novel end-fittings, the left end-fitting and the right end-fitting. Fig. 1a clearly shows that the left end-fitting is shorter and lacks the pneumatic arrangement as compared to the right end-fitting. The three primary functions of the end-fitting include; allowing the flow of compressed air into the PMA, maintaining the air-tight seal, and transferring the force generated by the PMA to the point of actuation in the application. The left and right end-fittings (Fig. 1b and c) have a few design features that are similar: the cylindrical portion, the raised ring portion, and the mounting arrangement. The end-fittings are attached to the two ends of the bladder-braided sleeve pair (Fig. 1a) to form an air-tight chamber within the bladder.

The assembled bladder encloses the raised ring and cylindrical portion of the two end fittings. As the bladder envelopes the cylindrical portion, the cylindrical portion is made of a diameter slightly larger than the bladder's inner diameter, which ensures that the bladder remains tightly in place and helps to form an air-tight seal. The clamps are fixed to the end-fittings at the cylindrical portion keeping the bladder-braided sleeve pair securely in place within the PMA. The length of the cylindrical portion is slightly larger than the

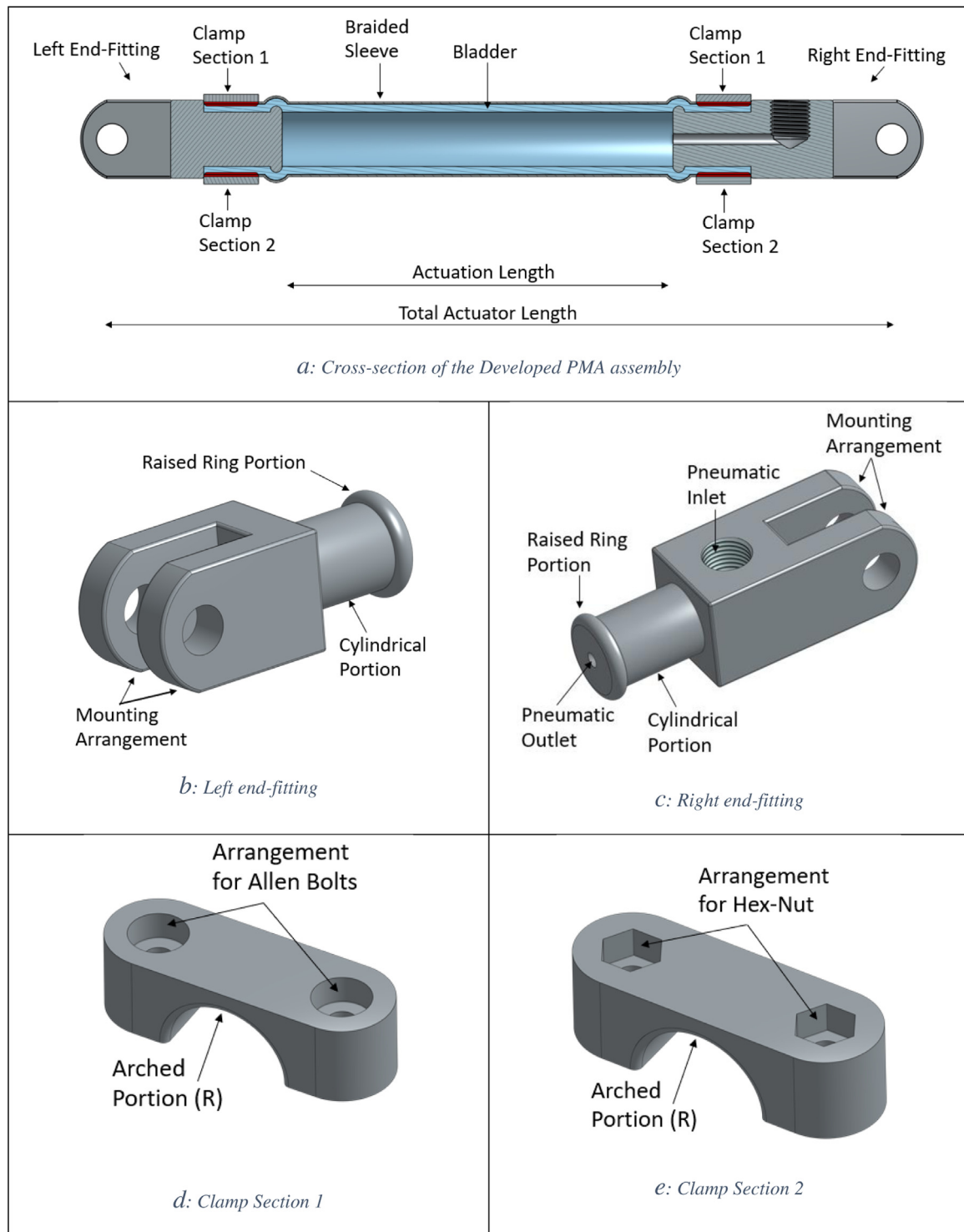


Fig. 1. The Developed PMA with the Novel End-Fittings and Clamps.

clamp's width to allow the clamps to fit over the end-fittings correctly. The raised ring portion of the end-fittings is another integral part of the end-fittings clamping assembly. The raised ring portion is designed to have a slightly larger diameter than the arch diameter ($R + R$) formed by the assembled clamp sections 1 and 2 (Fig. 1d and e), thus preventing the clamps from slipping off the cylindrical portion of the end-fittings. The significance of the cylindrical and raised ring portion is further highlighted in the clamp design. Another common feature to both the end-fittings is the mounting

arrangement, which consists of two legs with through holes to be attached via a pin connector. The PMA is attached to the mounting bracket (e.g., attached to the exoskeleton) at the gap between the two mounting legs of the end-fittings forming a pin joint with one rotational degree of freedom about the pin axis. This single degree of freedom at the pin joint facilitates an unrestricted actuation of the PMA.

The design feature which shows a variation in features of the left end-fitting from its right counterpart is the pneumatic arrangement

Table 1
Comparing Length of Developed PMA with the Commercial PMA used in this study.

Sr. No.	Pneumatic Muscle (PMA)	Actuation Length (mm)	Overall Length of PMA (mm)
1	Commercial PMA	100	260
2	Developed PMA	100	215

(Fig. 1c). The pneumatic arrangement includes a threaded hole for attaching the pneumatic coupling, which connects the PMA to the compressed air supply. The compressed air entering the pneumatic coupling is led through a path running axially through the cylindrical portion and exits beyond the raised ring portion into the air-tight chamber within the bladder (Fig. 1a). Combining the mounting, pneumatic, and bladder fixing arrangement into a single structure makes this a novel end-fitting (Fig. 1c) a compact and unique design. The provision for connecting the pneumatic coupling at the side face of the end-fitting provides enough room for the mounting arrangement. The second important feature of the novel end fitting design is the absence of the pneumatic arrangement in the left end-fitting, which reduces the overall length of the PMA without compromising the amount of contraction. The maximum contraction percentage achieved by the PMA lies in the range of 25–30 % (obtained experimentally) of its actuation length (Fig. 1a). Using a shorter actuation length to accommodate the PMA in a space-constrained application (e.g., exoskeleton) would significantly affect the amount of contraction. This loss of contraction length is seen majorly while using commercial PMAs, which come in standard actuation lengths. A shorter commercial muscle uses the same end fittings with the pneumatic arrangement at both ends with a shorter actuation length, thus producing a smaller amount of contraction. Unlike most commercial PMAs with the pneumatic arrangement at both ends, the PMA presented in this paper has the pneumatic arrangement in only the right end-fitting. This design of having the pneumatic arrangement only at one end significantly reduces the overall length of the PMA without compromising the amount contraction which depends on the actuation length as shown in Table 1.

Clamps are very crucial in the design of the end-fittings. The clamps form an integral part of the end-fittings, which secure the bladder-braided sleeve pair to the cylindrical portion of the end-fitting even at high loads and actuation pressures. The clamps designed in the present work consist of two sections labelled '1' and '2' (Fig. 1d and e). The common feature in the design of the clamp sections 1 and 2 is the arch with radius R , which is positioned at the cylindrical portion of the end-fittings over the assembled bladder-braided sleeve pair. The radius R is made slightly larger than the end fitting's cylindrical portion to compensate for the thickness added by the enveloping bladder and braided sleeve. The arch radius R is kept slightly smaller than the radius of the raised ring portion of the end-fittings, thus preventing clamps from slipping off the cylindrical portion of the end-fitting. The two clamp sections, '1' and '2', are positioned diametrically opposite the end fitting's cylindrical portion, thus providing an adequate clamping force that can be increased or decreased with bolts and nuts holding the two sections together securely. The interference fit between the arch portion of the clamps and the raised ring portion of the end-fittings results in a higher clamping force proportional to the actuation pressure and payload, thus making the PMA design much safer at higher loads. The shape of the holes is different at the clamp sections 1 and 2 (Fig. 1d and e). The clamp section 1 has equally spaced hexagonal-shaped holes on its flat face, which houses the hex-nut. The clamp section 2 has circular-shaped holes on its flat face, which allows the Allen bolt head to fit flush with the face. These Allen nuts and bolts are used to secure the clamps in place and prevent any accidental disassembly of the PMA.



Fig. 2. Assembled PMA of weight 90gms.

The primary assembly constituting of the PMA using the novel end-fittings is a quick and straightforward process. The first step in the assembly of the PMA involves inserting the flexible bladder into the braided sleeve. The next step involves attaching the end-fittings at either end of the bladder-braided sleeve assembly, which is done by sliding the bladder and sleeve over the raised ring and cylindrical portion of the end-fittings. The last step involves fixing the clamps over the assembled bladder-braided sleeve pair at the cylindrical portion of the end fittings. The clamps are then tightened in place using Allen bolts and nuts. Fig. 1a clearly shows the assembled PMA, wherein the cylindrical portion is enclosed by the bladder-braided sleeve pair and sandwiched between the two sections (1 and 2) of the clamps.

The PMA developed in this study was fabricated using a commercial bladders and braided sleeves to bring in ease of component availability, as shown in Table 2. A commercially available styrene-based thermoplastic elastomer tube was used as the bladder, which has a tensile modulus of 0.84 MPa (4×6 , 6×9 , and 12×17 bladders) and 0.89 MPa (for 10×14 bladder). These modulus values closely match the properties of the skeletal muscle [22] when compared to natural or Silicon-based Rubber [23]. A commercial polyethylene terephthalate expandable cable-manager having a tensile strength of 416 MPa was used as the braided sleeve. The length of the PMA considered in this study is 100 mm, which is equivalent to the commercially available PMA used in this study (comparative analysis in section 3.5). The end fittings and clamps which hold the bladder-sleeve assembly are 3D printed using Polylactide (PLA), which is structurally strong, lightweight, and biodegradable [24]. The performance of the PMA is studied for various bladder and braided sleeve sizes, as shown in Table 2. The photo of the assembled 10×14 PMA is shown in Fig. 2, whose total weight is 90gms.

3. PMA characterization

The characteristics of the PMA, such as the force versus contraction ratio and deflection of the PMA for various pressure and endurance, are studied to establish its use in the intended application, which is the elbow exoskeleton in this case. The four developed PMA's, along with one commercially available PMA,

Table 2
Dimensions of bladder and braided sleeve used in the developed PMA.

PMA Name (size IDxOD (both in mm))	Thickness of Bladder Tube(mm)	Tensile strength of tube MPa*	Tensile Modulus of tube MPa*	Diameter of braided sleeve (mm) Used
12 × 17	2.5	4.54	0.84	19.05
10 × 14	2	7.75	0.89	19.05
6 × 9	1.5	4.54	0.84	9.53
4 × 6	1	4.54	0.84	6.35

* Experimentally determined values.

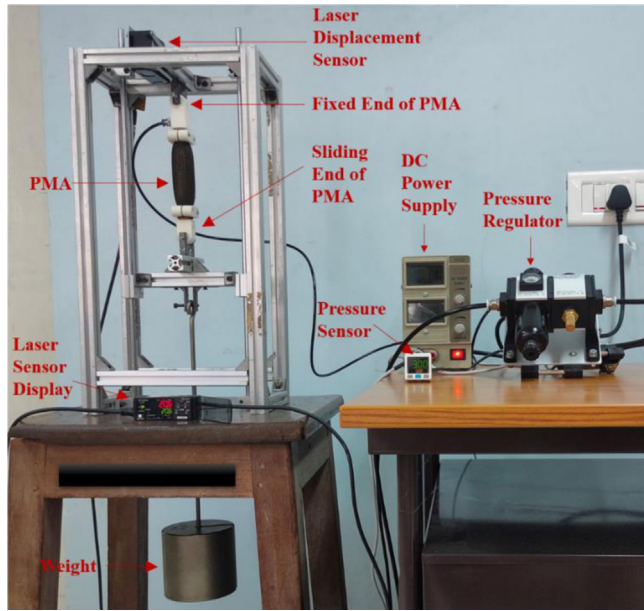


Fig. 3. Photograph of the PMA test rig.

are experimentally tested and presented in this study. The subsequent subsection details the test setup for characterizing the PMA. The Force and Contraction characteristics study gave an overall understanding on the inverse relation between force generated and amount of contraction, which was conducted for muscles of varying diameters. The Pressure vs Deflection and Pressure vs Force tests were more specific tests in order to understand the maximum amount of deflection and force achieved by the developed muscle at different operating pressures. These tests also help in understanding hysteresis and repeatability errors at various pressures in the developed muscle and a commercially available PMA. The tests of Force vs Contraction and Pressure vs Force and Deflection were carried out at quasi static conditions.

3.1. PMA test rig Development

The developed pneumatic muscle is intended to be used for medical rehabilitation applications. This calls for testing and establishing the performance of the PMA before using it on patients. The exoskeleton for rehabilitation of the upper extremity has a maximum load of 5Kg [25,26]. Hence, the maximum load used for testing is restricted to 10Kg, keeping a Factor of Safety of 2. The characterization involves testing parameters such as force, deflection, and endurance. Hence, a simple but modular test-rig that facilitates all tests with little modification to the test-setup is necessary [27]. Aluminium t-slot sections of size 20 mm x 20 mm are used to fabricate the test setup due to their lightweight and modular nature. The setup for experimental testing of PMA is shown in Fig. 3. The outer frame consists of 12 modular sections, forming a cuboid of dimensions 460 × 235 × 150 (H x W x B in mm). The structure's top face

has an aluminium section parallel to the breadth, which houses a clamp to mount the pneumatic muscle actuator's fixed end. A laser displacement sensor (Keyence IL300) having an accuracy of 0.1microns is used to measure PMA deflection and is mounted on the top face of the test setup. The free end of the PMA is attached to a sliding mechanism of the test rig. The sliding mechanism consists of two linear bearings on either side of the structure, which ensures smooth sliding of the pneumatic muscle. The PMA is tested by suspending dead weights on the sliding mechanism of the test rig. These suspended weights exert the required force on the free end of the PMA.

The PMA is actuated by supplying compressed air from the compressor. The air pressure is regulated by a proportional pressure regulator having a pressure range of 0.05–10 bar to control the extent of actuation. A digital pressure sensor (P43 Series from Metal Works) records the system's pressure and serves as a feedback device in the life cycle test. Section 3.2–3.6 details the various tests performed on the developed and commercial muscles.

3.2. Force versus contraction characteristics

The experimental testing is performed to understand the extent of force and contraction the pneumatic muscle produces for a particular value of pressure. For a particular pressure, the PMA produces a maximum force when both of its ends are fixed, which is known as blocked force, while it produces maximum deflection if only one end is fixed. In this test, the pressure is maintained constant for every trial (e.g., 0.5R is the retraction stroke at 0.5 bar pressure). Each trial involves an incremental rise in the force at the free end (or sliding end) of the PMA, which was imparted by hanging dead weights. The variation in contraction percentage with increasing force was observed and presented in the plots shown in Fig. 4f–Fig. 5k. Each trial was conducted up to weight (equivalent force) of 10Kg, which was increased incrementally at a rate of 0.5Kg per contraction percentage of the PMA observed. Subsequently, the experiment involved increasing the pressure by 0.5 bar for every trial (extension(E) and retraction(R) cycle), up to a pressure of 4 bar. The 4 × 6 PMA was tested up to a pressure of 3.5 bar on account of its smaller bladder volume.

The plots shown in Fig. 4f–Fig. 5k indicate that for a particular value of pressure, the amount of force is inversely proportional to the contraction percentage; that is, as the force increases, the contraction percentage decreases. The smaller muscles 4 × 6 and 6 × 9 PMAs have a more considerable contraction variation, with an increase in force for a particular actuating pressure. In comparison, the larger cross-section muscles (10 × 14 and 12 × 17) have a steeper slope (smaller variation in contraction with increasing force). This behavior is mostly due to the effect of the hyperelastic bladder, whose volume is more prominent in smaller muscles (as shown in Table 3). Therefore, PMA's of larger cross-sections tend to have better positional stability (and accuracy) and are less affected by variation in applied load.

Table 3 depicts the ratio of bladder volume (V_{bladder}) and the total volume of the actuator (V_{actuator}). The bladder is made of a hyperelastic material and constitutes a large volume portion

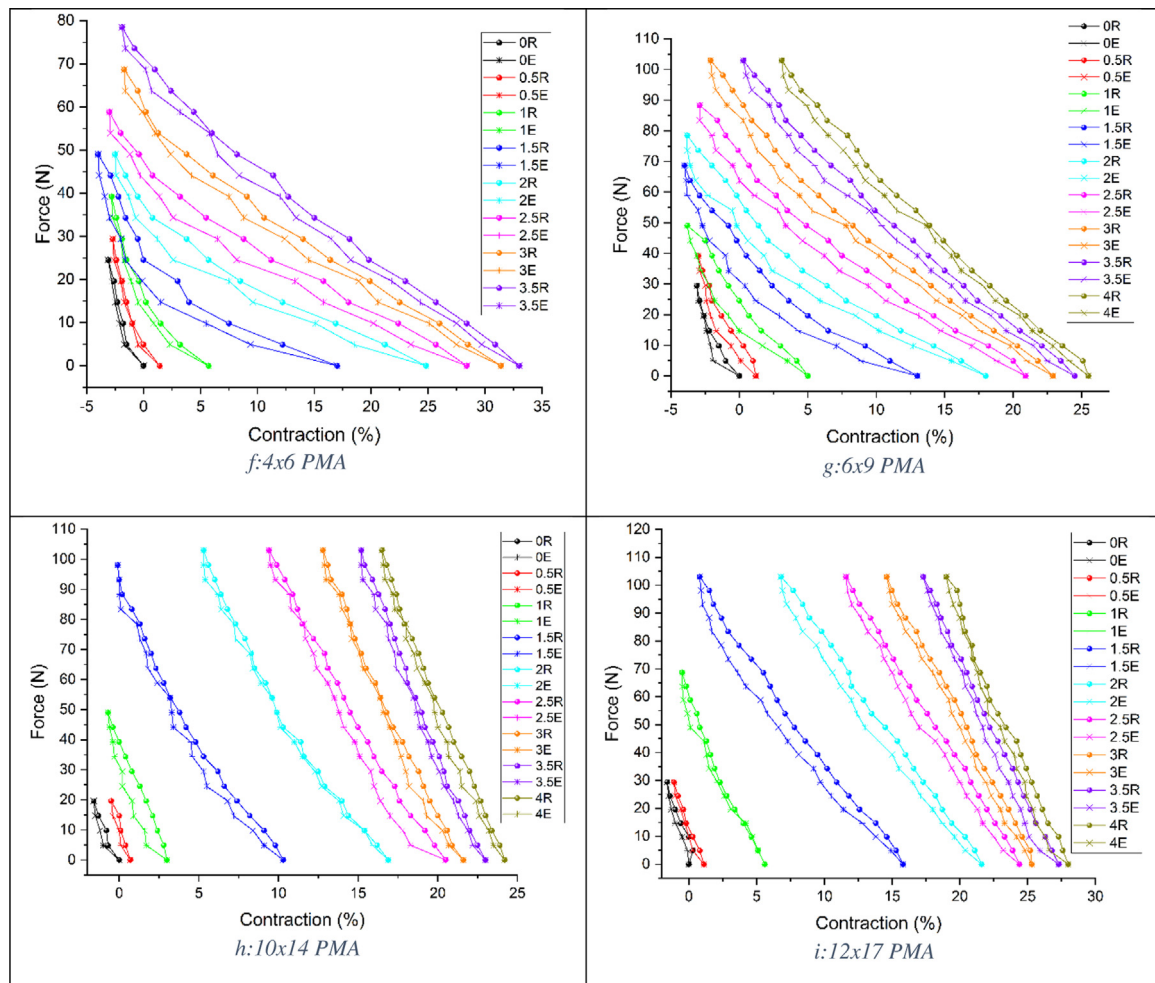


Fig. 4. Force vs. Contraction Ratio of Developed PMA (Note: 1R = Retraction Stroke at 1 bar; similarly, 1E is Extension Stroke at 1 bar).

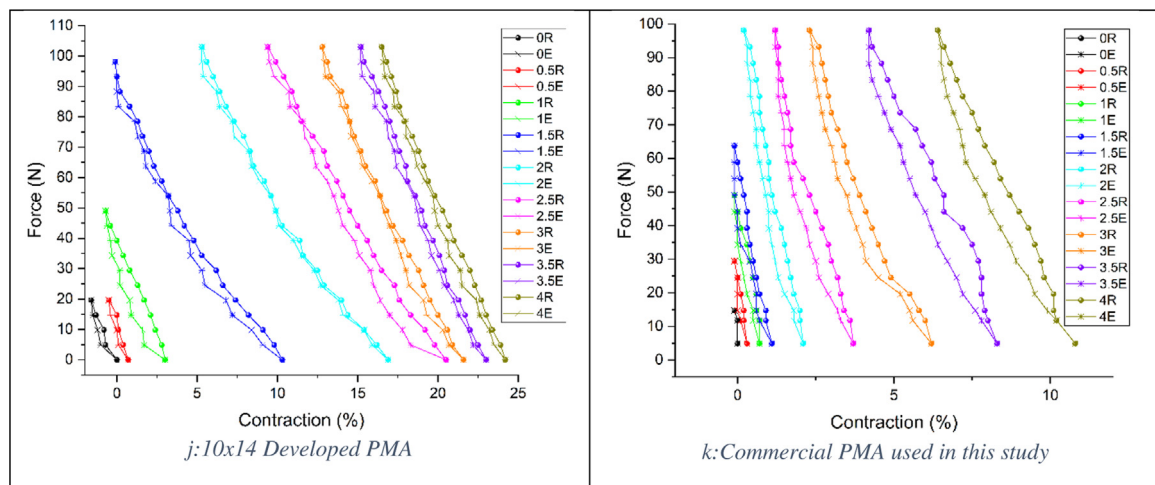


Fig. 5. Force vs. Contraction Ratio of Developed and Commercial PMA of same diameter and actuation length.

Table 3

Volume of bladder w.r.t. the actuator volume.

PMA	V _{bladder} /V _{actuator}
12 × 17	0.89
10 × 14	0.73
6 × 9	0.94
4 × 6	0.94

of the entire pneumatic muscle actuator; thus, its characteristics profoundly affect the behavior of the overall PMA. An important observation from Table 3 is the least volume ratio is found in the 10 × 14 muscle; thus, it tends to be least affected by the effect of the hyperelastic bladder. Therefore, the 10 × 14 muscle has better positional accuracy, low hysteresis error (higher repeatability of force or deflection achieved for a particular actuation pressure), and is least

affected by increasing payload. The 12×17 PMA (Fig. 4i) shows a greater fall in contraction with the increase in force in comparison to the 10×14 muscle (Fig. 4h), resulting from the 10×14 muscle having a bladder of slightly higher tensile modulus and tensile strength (Table 2).

The variation in contraction with increasing force or load is the compliance of the pneumatic muscle, which results from the hyperelastic bladder and the use of a compressible fluid (air). The restriction to the motion of the PMA results in a certain percentage of elongation of the hyperelastic bladder, which returns to its set position (based on applied pressure) once the restriction is removed. Compliance is an essential property required in a rehabilitation exoskeleton, which prevents harm to the patients who at times offer resistance to the exoskeleton's motion [28]. The plots (in Fig. 4f–i) indicate the variation of contraction percentage vs. force, which is directly dependent on the bladder's size and material properties. At higher values of contraction, the flexible inner tube is being strained beyond the linear stress-strain relation, resulting in a stiffness effect, which is evident in the smaller PMAs compared to its larger counterparts [8].

The plots also reveal a hysteresis error during the retraction-extension (R-E) strokes in both the developed ($4 \times 6 \times 9$, 10×14 , and 12×17) and the commercially available muscles. Several factors contribute to hysteresis, which includes the hyper-elastic nature of the bladder, friction between braids, and friction between braided sleeve and bladder. The plot exhibiting the least hysteresis is the 10×14 (Fig. 4h) muscle, resulting from the smaller bladder volume and a larger gap between the bladder and braided sleeve. All these factors prove that the 10×14 pneumatic muscle is best suited for the rehabilitation application, as it exhibits a perfect balance of compliance and positional accuracy.

The Plots in Fig. 4f–Fig. 5k show that for an actuation pressure of 4 bar, the PMA developed in the present work contracted by about 30 %, while the commercial muscle contracted by only 11 % of its actuating length. This paper compares the behavior of the 10×14 PMA (Fig. 5j) with the tested commercially available muscle (Fig. 5k) due to their similar dimensional properties (14 mm diameter, 2 mm thickness, and 100 mm length). The plots shown in Fig. 5j and k reveal that the 10×14 muscle follows a similar behavior (of force vs. contraction) as the commercial muscle. The plots shown in Fig. 4f–Fig. 5k reveal the least hysteresis error is exhibited in the 10×14 muscle, which uses a tube of the least volume ratio having a tensile modulus slightly lower than the rest of the developed pneumatic muscles. The force and contraction behavior of both 10×14 (Fig. 4h) and 12×17 (Fig. 4i) seems very similar, as they use a thicker tube and use the same size of the braided sleeve (same actuator volume). This test clearly shows the variation in the behavior of force and contraction for different PMA sizes. The plots in Fig. 4f–Fig. 5k show an increase in the amount of force generated for a particular value of pressure with the increasing size of the PMA. The plots also indicate that the variation of contraction with force is higher for the smaller cross-section muscles than the larger muscles. It can be concluded that the sizes 10×14 and 12×17 produce a stable force and contraction with incremental pressures.

3.3. Pressure vs deflection characteristics

The force versus contraction ratio is used to get the overall behavior of the various pneumatic muscle's tested. In assistive applications such as an elbow exoskeleton, it is essential to accurately control the position of the PMA. The pressure versus deflection behavior provides a good basis for deciding the control strategy to actuate the pneumatic muscle. It is established from the force versus contraction study that the 10×14 PMA has high positional accuracy and a low hysteresis error. Accordingly, the 10

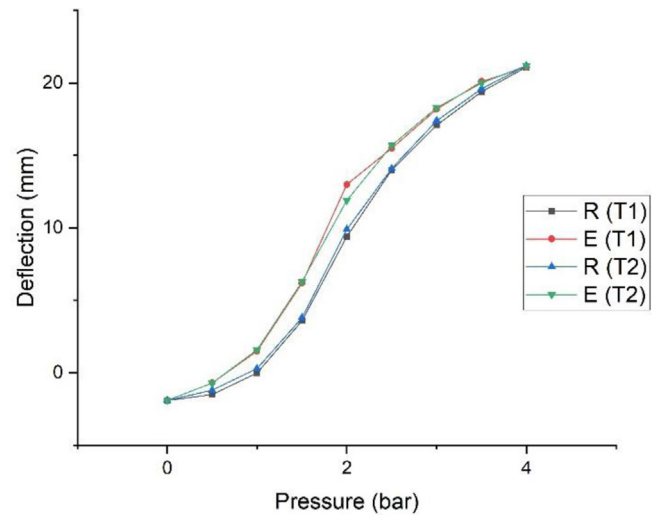


Fig. 6. Deflection vs. Pressure of 10×14 PMA (R(T1): Retraction Stroke of the 1st Trial, E(T1): Extension stroke of the 1st trial).

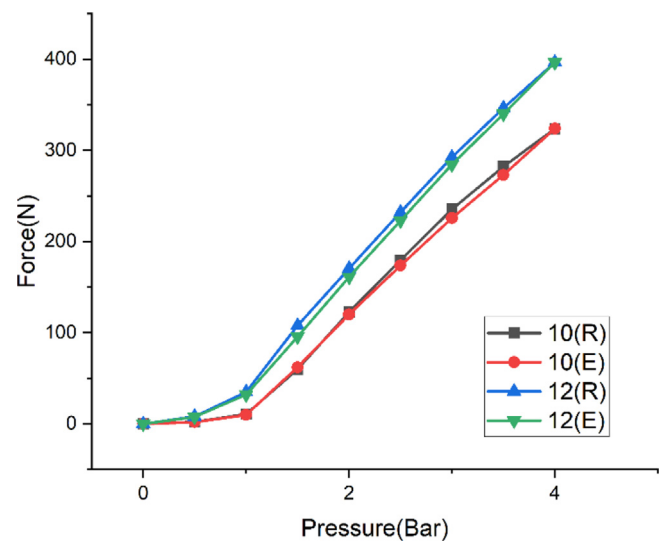


Fig. 7. Force vs. Pressure of 10×14 (10R & 10E) and 12×17 (12R & 12E) PMA for Extension(E) and Retraction (R) Strokes.

$\times 14$ PMA is compared with the commercial muscle in the pressure versus deflection test. As shown in Fig. 3, this test involves mounting one end of the pneumatic muscle at the test rig's top fixed support while clamping the other end to the sliding assembly, which was loaded with a 5Kg dead weight. As previously mentioned, the maximum load while actuating the forearm through flexion is less than 5Kg, which includes the weight of the support structure, patient's hand, payload, and the patient's forearm itself [25,26]. In this test, the laser displacement sensor is used to measure the deflection (retraction/extension) of the muscle. The laser displacement sensor records the amount of deflection experienced by the loaded PMA for incremental pressure input. The test was conducted in pressure steps of 0.5 bar up to a pressure of 4 bar for the 10×14 PMA.

Fig. 6 shows a linear deflection within the pressure range of 1–3 bar, while the curve tends to flatten out at its ends. This trend indicates a linear relationship between pressure and deflection from 1 bar to 3 bar, thus resulting in the ideal control range for the 10×14 muscle. The flattened portion at the start of the plot is a result of the gap between the bladder and braided sleeve (pressure dead-band), while that at the end is most likely due to the expanded

bladder reaching its maximum radial limit. The presence of a hysteresis error visible in Fig. 6 indicates that it is difficult to achieve the same value of deflection during the retraction (R) and extension (E) strokes of the PMA.

The plot in Fig. 6 showcases two trials (T1 and T2) to check for the repeatability of the actuators. It is evident from both the plots that there is a slight difference in the deflection between trial T1 and T2 for a particular value of pressure, resulting from the hyperelastic nature of the bladder. The non-linear nature of pressure and deflection is a result of friction (between braids of the braided sleeve and braided sleeve & bladder) and the bladder's hyperelastic nature. Therefore, feedback control is mandatory to control the position of the PMA accurately.

3.4. Pressure vs force characteristics

The force generated by the PMA is an essential parameter in deciding the size of the pneumatic muscle to be used based on the requirement of the application. For instance, the PMA used for an upper limb exoskeleton requires to generate a large force in order to flex the elbow (5Kg payload) of the patient. The test was conducted with few modifications to the test rig shown in Fig. 3. The sliding mechanism shown in Fig. 3 is fixed to the base of the test rig forming a rigid base. A load cell (HBM 2 KN) was attached to the free end of the PMA, and the other end of the load cell was anchored to the rigid base of the test rig. The force data from the load cell was acquired using a DAQ (NI9237 4 channel strain gauge DAQ). The pneumatic muscle was actuated using the proportional pressure regulator, and its force was recorded for various values of pressure. The Force vs. Pressure behavior of two developed muscles, 10×14 and 12×17 (Fig. 7) were tested to evaluate the effect of size on the muscle Force behavior.

Fig. 7 clearly indicates that the larger diameter muscle 12×17 produces a higher force of 401 N while the 10×14 generates a blocked force of 324 N at 4 bar actuation pressure. Similar to the deflection characteristics the value of force for a particular pressure in the retraction(R) and extension(E) strokes is different, thus indicating a hysteresis error. Fig. 7 also indicates that the 10×14 PMA showed a smaller hysteresis error in comparison to the 12×17 PMA, which is due to the higher tensile modulus bladder used in the 10×14 PMA. The plots (Fig. 7) show a linear relation between force and pressure in the pressure range of 1–4 bar. The delay in the force generation after application of pressure (<0.25 bar) is due to the pressure deadband (or gap between the bladder and braided sleeve). The plots also indicate a smaller pressure deadband in 12×17 muscle (smaller gap between bladder and braided sleeve) as a result of which it starts producing force at a pressure less than 0.25 bar when compared to the 10×14 PMA (deadband~0.5 bar).

3.5. Comparing deflection and force characteristics of the developed PMA with a commercial PMA

The commercial McKibben muscle forms an excellent standard to compare the performance of a newly developed PMA. The commercial PMA used in this test has the dimensional properties of the 10×14 muscle with 10 mm internal diameter, 14 mm outer diameter, and a 100 mm actuation length. The displacement and force tests were conducted up to the maximum operating pressure of 8 bar for the commercial muscle.

The Fig. 8l shows the relation of muscle deflection with increase in actuation pressure. Similar to the 10×14 PMA presented in the section 3.3 the commercial muscle shows a linear relation between pressure and deflection in the pressure range of 2 bar–6 bar. The Fig. 8l also indicates a flattening at the start and end of the plot which is a result of pressure deadband and the bladder reaching its maximum limit of expansion. From Fig. 6 and Fig. 8l, it is clear

that the developed muscle (10×14) produces a higher deflection at a lower pressure when compared with the commercial muscle. The plots also indicate a smaller hysteresis error in the developed PMA (10×14) in comparison with the commercial muscle.

The Fig. 8m is the plot of Pressure versus Force for the commercial PMA. The Force behavior of the commercial PMA produces approximately 350 N at 8 bar, shown in Fig. 8m. The plot in Fig. 8m also shows the flattening of the force curve at its ends, thus indicating an initial pressure deadband and a negligible increase in the force with the increase in pressure beyond 7 bar. The developed PMA in Fig. 7 shows a linear increase in force with pressure in the operating range of 1 bar–4 bar while a similar trend is observed in the range of 2 bar–6 bar in the commercial PMA (Fig. 8m). The force generated by the commercial PMA (Fig. 8m) at 4 bar is under 150 N, which is much less than the force generated by the 10×14 developed PMA (Fig. 7) having a similar geometric configuration. The plots in Fig. 7 and Fig. 8m, therefore, indicate that the developed PMAs using the styrene-based bladder and PET braided sleeve are able to generate a higher force at a lower actuation pressure therefore requiring a smaller compressor, thus reducing the cost of the system.

3.6. Life cycle test

As the developed pneumatic muscle is intended to actuate an exoskeleton for medical rehabilitation, its reliability is critical. To test the reliability of the PMA before actually using it on patients, an endurance test or life cycle test is designed and implemented. The test was conducted by mounting the muscle onto the test rig (Fig. 3) and loading it with a 5Kg load (actual load of the structure, forearm, hand, and payload) at its free/sliding end. Unlike the previous tests, this test involved a few additional components, such as a microcontroller (Atmega 2560 clocked at 16 MHz), a pressure sensor (P43 Series from Metal Works), and an electro-pneumatic pressure regulator (Regtronics ¼ from Metal Works). The microcontroller is programmed to give a cyclic voltage signal (incremental then decremental) to the pressure regulator to increase and decrease the pressure of compressed air fed to the pneumatic muscle in order to realize the contraction and extension strokes of the PMA. The pressure sensor connected to the system senses the system pressure and acts as a feedback device for the controller in case of a sudden pressure drop, indicating a rupture (failure) of the pneumatic muscle. The entire program was run in a loop, and the controller recorded the number of successful cycles the PMA endured. The flow chart in Fig. 9 gives a detailed overview of the lifecycle test. The life cycle test was conducted for the developed 10×14 muscle with a 5Kg load attached which ran for over 200000 cycles over 8.38 days. The test was conducted as close to real conditions as possible considering the load attached (5Kg) and the maximum operating pressure (4 bar for 10×14) fed at each cycle to realize a full contraction stroke.

4. Mathematical model of a pneumatic muscle actuator (PMA)

4.1. Mathematical models of PMA

A mathematical formulation gives the theoretical perspective of the pneumatic muscle in an ideal scenario. Several authors have reported on the modelling of pneumatic muscles to simulate and optimize the PMA design for different applications. A comparative study on the various mathematical models is presented in this section to establish a model which resembles the fabricated pneumatic muscle. This study helps in the process of optimization of a

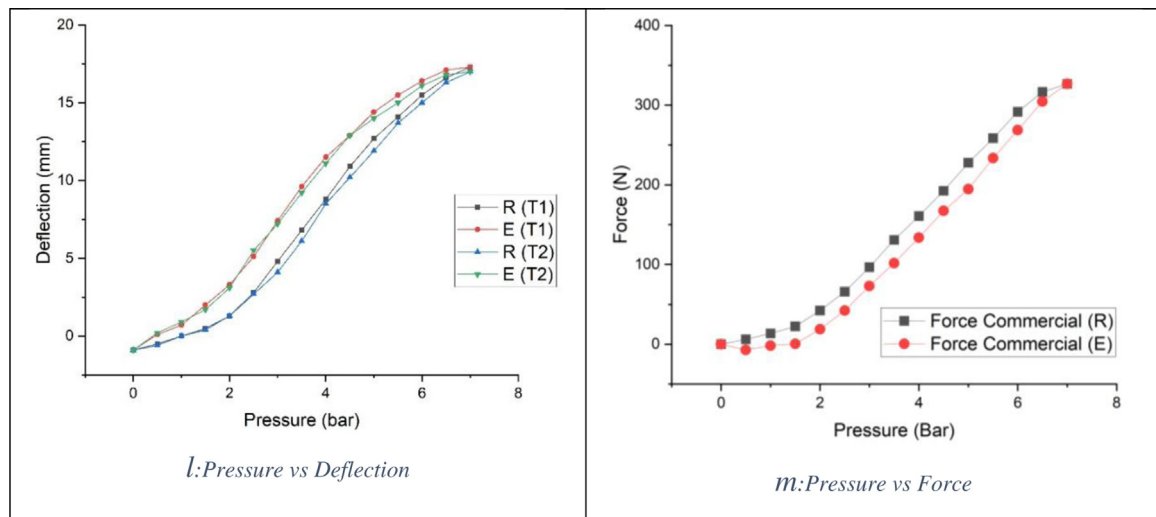


Fig. 8. Deflection and Force Characteristics of the tested commercial PMA.

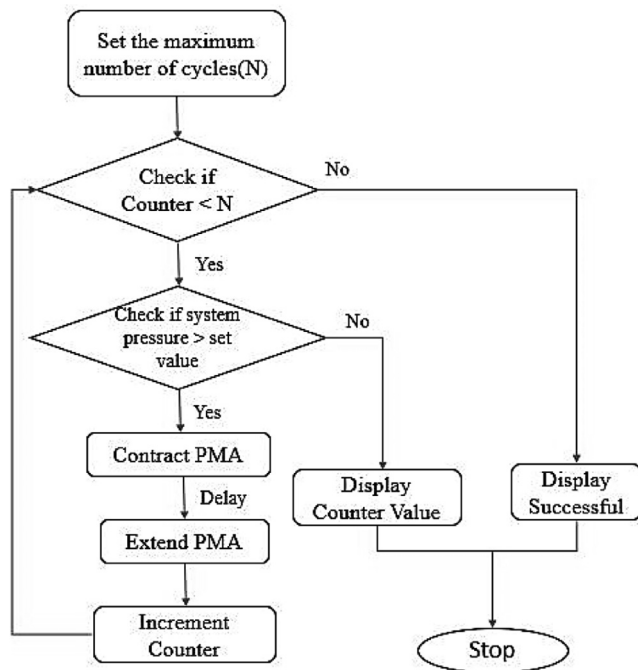


Fig. 9. Lifecycle Test Flowchart.

pneumatic muscle and saves both time and resources, which is a downside of a purely experimental approach.

The PMA's volume substantially comprises the hyper-elastic bladder (e.g., latex rubber), thus making the actuator highly non-linear and is therefore challenging to model. Fig. 10 shows a single braided fiber wrapped around the bladder at an initial angle of ' α_0 ' with respect to the central axis of the muscle. 'F' is the force the pneumatic muscle produces on the application of pressure 'P.'

The actuation of the pneumatic muscle involves the application of compressed air at a set pressure 'P', which causes the outward expansion of the bladder. The outward motion of the bladder is confined by the braided sleeve, which in turn converts the radial motion to axial contraction of the muscle. The axial contraction is expressed by the contraction ratio.

$$\text{Contraction Ratio} = \frac{\text{Length of PMA after actuation (L)}}{\text{Original length of PMA (L}_0\text{)}} \quad (1)$$

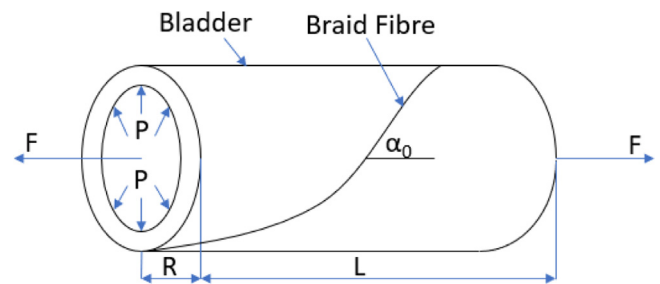


Fig. 10. Schematic of PMA with braid fiber (F in X direction).

Where $L_0 > L$

Length of PMA after actuation (L) can be obtained by deducting the amount of deflection (δ) from the original length of the muscle (L_0):

$$L = L_0 - \delta \quad (2)$$

When the pneumatic muscle's motion is restricted by clamping one end and loading the other, it tends to apply a force on the load, which is proportional to the applied actuation pressure. This actuation force the pneumatic muscle produces for an applied pressure was first presented by Gaylord [3,10,7] as depicted in Eq. 3

$$F_{\text{Gaylord}} = \frac{P^* (3 * L^2 - B^2)}{4 * N^2 * \pi} \quad (3)$$

Where P is the pressure of compressed air applied to the PMA

N is the number of turns of a single braid fiber about the bladder

L is the instantaneous length of the actuator

B is the length of a single braid fiber

The diameter of the flexible bladder and the braid angle plays a vital role in the amount of contraction and force generated by the pneumatic muscle. The formulation given by Gaylord has not accounted for parameters such as friction, the effect of braid, and bladder. The effect of braid and bladder on the behavior of the PMA was introduced in the formulation given by Tondur [3].

The force (F_{M1}) formulation by Tondur [3], which includes Gaylord's Force, is given by Eq. 4. This model is represented as M1 in this paper.

$$F_{M1} = F_{\text{Gaylord}} - \frac{\pi * R_0^2 * P * (2 * h_0 * (2 * \cos \alpha_0^2 * (1 - \delta)^2 - 1))}{(1 - \delta) * (1 - \cos \alpha_0^2 * (1 - \delta^2))} \quad (4)$$

Where α_0 is the initial braid angle ($\alpha_0 = 25.3^\circ$)

R_0 = initial radius of the tube ($R_0 = 7$ mm)

t_0 = initial thickness of the bladder ($t_0 = 2$ mm)

$h_0 = t_0/R_0$

The bladder is made of a hyperelastic material (styrene-based thermoplastic elastomer), an essential parameter while modeling the pneumatic muscle. Kothera et al. [7] presented two models; the first model (M2) considers the hyperelastic nature of the flexible bladder, and the second model (M3) based on Newton's second law.

The model by Kothera et al. [7], which accounts for the hyper-elastic bladder, comprises two primary elements. The first is the Gaylord Force term, and secondly, a corrective term for the hyper-elastic bladder. The model also takes into consideration the effect of the braided sleeve on the force generated by the muscle.

The Mooney Rivlin Force ($F_{\text{hyperelastic}}$), as given in Eq. 5, takes into account the hyper-elastic nature of the bladder.

$$F_{\text{hyperelastic}} = 2 * V_{bl} * [C_{10} * (\lambda_1 * \lambda_1' + \lambda_2 * \lambda_2' + \lambda_3 * \lambda_3') + C_{01} * (\lambda_1 * (\lambda_2^2 + \lambda_3^2) * \lambda_1' + \lambda_2 * (\lambda_3^2 + \lambda_1^2) * \lambda_2' + \lambda_3 * (\lambda_1^2 + \lambda_2^2) * \lambda_3')] \quad (5)$$

Where C_{01} and C_{10} are Mooney Rivlin constants ($C_{01} = 0.0594$; $C_{10} = 0.00306$) obtained experimentally for the Thermoplastic elastomer used for the bladder, and λ_1 , λ_2 , and λ_3 are the stretch ratios given as $\lambda_1 = L/L_0$, $\lambda_2 = (R - t)/(2R_0 - t_0/2)$, and $\lambda_3 = t/t_0$. Where 't' is the instantaneous thickness of the bladder

Assuming a bladder of constant volume, the volume of the bladder (V_{bl}) is given by Eq. 6

$$V_{bl} = L_0 * [\pi * R_0^2 - \pi * (R_0 - t_0)^2] \quad (6)$$

Where,

L_0 is the initial length of the PMA

t_0 is the initial thickness of the bladder

The change in radius (R) of the bladder after application of pressure is given by Eq. 7

$$R = \frac{\sqrt{B^2 - L^2}}{2 * \pi * N} \quad (7)$$

The braided sleeve plays an important role in the behavior of the PMA; therefore, considering it as an integral part of the muscle model is essential. The braided sleeve is modeled using Eq. 8.

$$F_{\text{braid}} = V_{br} * \left(\frac{4 * \pi^2 * p^2 * B^2 * L}{E_{br} * A_{br} * n^2 * (2 * \pi * N)^4} \right) \quad (8)$$

Where V_{br} is the volume of a single braid fiber

E_{br} is the modulus of elasticity of the braid fiber (for the current work, a PET braided sleeve is used)

A_{br} is the cross-sectional area of the braid fiber

n is the number of braid fibers in the braided sleeve

Therefore, using Eq. 3, Eq. 5, and Eq. 8, the PMA force formulation (F_{M2}) can be written as Eq. 9:

$$F_{M2} = F_{\text{Gaylord}} - F_{\text{hyperelastic}} - F_{\text{braid}} \quad (9)$$

The third model (M3) presented in this paper, as proposed by Kothera et al. [7], is based on Newton's second law or using the force balance method. The forces in x (along the length of PMA) and z (radial to the bladder) directions shown in the PMA schematic (Fig. 10) are equated to get the force (F_{M3}) produced by the PMA in Eq. 10.

$$F_{M3} = F_{\text{Gaylord}} + P * \left(\frac{V_{bl}}{L} - \frac{t * L^2}{2 * \pi * R * N^2} \right) + E_t * V_{bl} * \left(\frac{1}{L_0} - \frac{1}{L} \right) - \frac{E_t * L}{2 * \pi * R * N^2} * (t * L - t_0 * L_0) \quad (10)$$

Where E_t is the tensile modulus of the bladder. In this work, a styrene-based thermoplastic elastomer bladder having a tensile modulus of 0.893 MPa (obtained experimentally) is used.

All the models discussed above were programmed in MATLAB (flowchart of complete procedure given in Appendix A) and simulated for various pressures and contraction ratios. In all three models, a constant bladder volume is assumed, and the gap between the braided sleeve and the bladder is accounted for by considering a pressure deadband of 0.2 bar (obtained experimentally). The comparative analysis of simulation with the experimental findings is presented in section 4.2.

4.2. Comparing experimental and simulated results

The use of a hyper-elastic bladder makes pneumatic muscle actuators non-linear in nature. Thus, it is essential to determine the mathematical model that best describes the developed muscle's behavior. This paper evaluated three mathematical models (M1, M2, and M3) of pneumatic muscles and compared their results to the experimental (Exp) results. The comparison was made for three different pressures, and the results are shown in Fig. 11n–p.

The plots reveal that as the pressure is increased, the experimental results tend to follow the simulated results more closely. This may be mainly due to the increased stiffness and reduction in the gap between the bladder and braided sleeve at increased pressures, thus demonstrating a close to ideal behavior. The plots in Fig. 11n–p make it clear that model M2 best represents the developed pneumatic muscle. As model M2 takes into account the hyper-elastic nature of the bladder and the effect of the braided sleeve, it has an accuracy higher in comparison to models M1 and M3. The accuracy of the simulation can be improved by adding a correction factor, but the main aim of this study was to find a model which closely resembles the force and contraction characteristics of the developed PMA.

5. Conclusions

In this work, an easy to assemble pneumatic muscle actuator is designed, developed, fabricated, and experimentally characterized. The highlight of the PMA designed is the novel end fittings, which are 3D printed and used along with a commercially available bladder and braided sleeve to prove the operational versatility. The design and development of a modular test rig helped to conduct all tests with minor adjustments to the test setup. The muscle characterization was carried out w.r.t. parameters such as pressure, contraction ratio, deflection, and force. The Force versus contraction ratio gave the overall performance of the pneumatic muscle. From this test, it is evident that the four developed muscles demonstrated a hysteresis error, but the amount of this error seemed to decrease in the larger diameter muscles (10×14 and 12×17). This study also compares the 10×14 pneumatic muscle with a commercial muscle that shows a similar behavioral trend of force and contraction with variation in pressure. The advantages of the developed muscle in comparison to the commercial muscle include:

- 1 The novel end-fittings simplifies the process of assembly and maintenance of the developed PMA. Thus, allowing the user the choice of bladder and braided sleeve that best suits their intended application. In this case as the application is an elbow exoskeleton therefore a styrene-based bladder was used whose tensile modulus is similar to a skeletal muscle.
- 2 The novel end-fittings also resulted in a PMA of smaller form factor without compromising the amount of contraction. 3D printing the end-fittings also helped to realize an overall lightweight and cost-effective PMA.

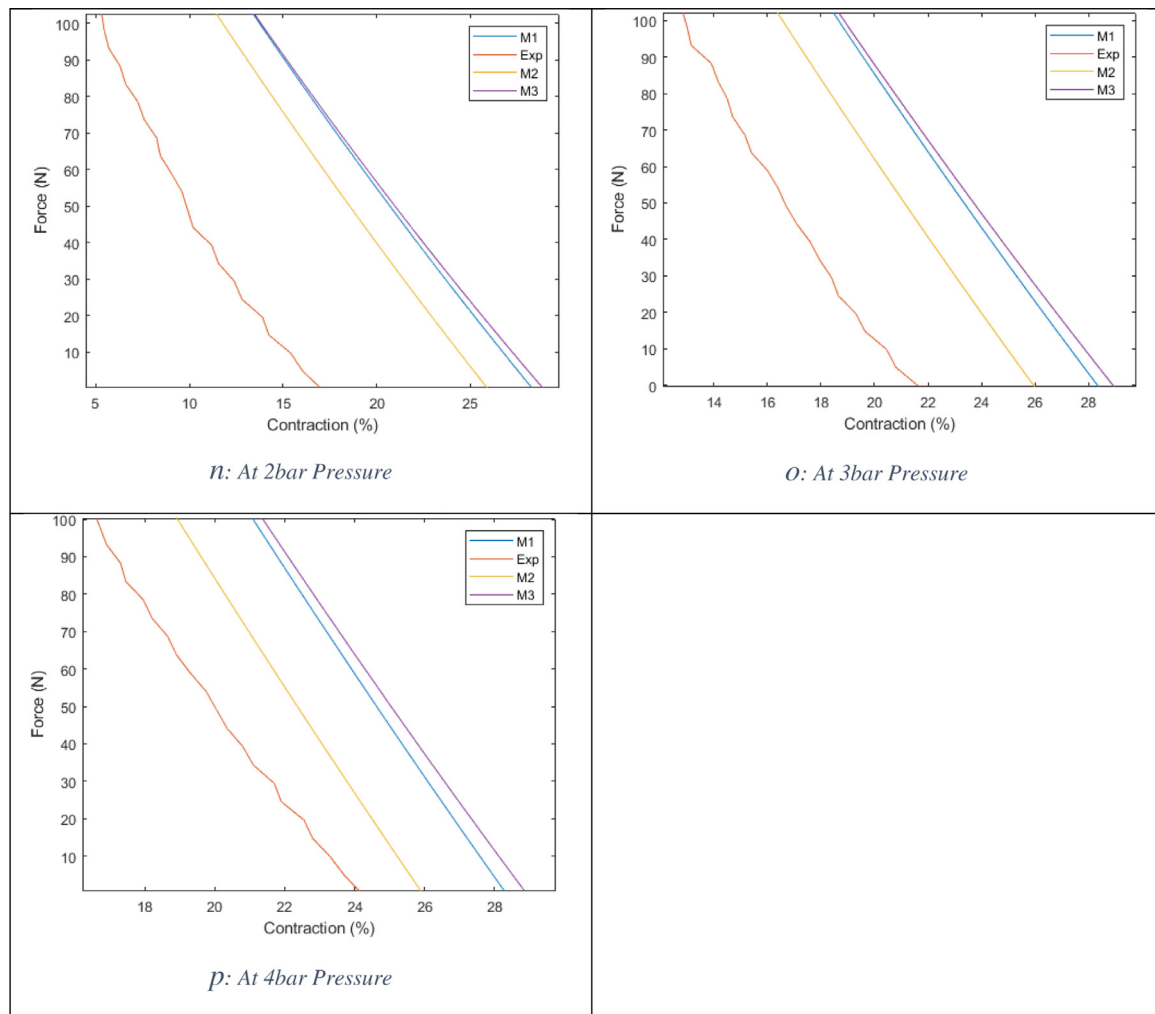


Fig. 11. Force vs. Contraction (Where 'M1', 'M2', & 'M3' are Mathematical Models and 'Exp' is obtained experimentally).

- 3 The developed muscles produced a contraction 18 % higher than the tested commercially available muscle at a 4 bar actuation pressure.
- 4 The developed muscles generate a force 50 % higher than the commercially available muscle tested at an actuation pressure of 4 bar. The developed muscles also demonstrated a linear increase in force with the increase in actuation pressure.
- 5 As the developed muscle works at lower pressure compared to the commercial muscle used in this study, it can be operated with a smaller power pack (compressor).
- 6 The 10×14 muscle demonstrated a smaller hysteresis error than the commercial muscle tested, which is most likely because the bladder occupies a smaller overall volume within the actuator than the other developed and commercial PMA tested.
- 7 The lifecycle test revealed a life of over 200000 cycles at actual loading (5Kg for elbow exoskeleton) and maximum operating pressure (4 bar) conditions.

Three mathematical models of the PMA were evaluated, and they were compared with the experimental results. Observations from the force versus contraction plots include:

- 1 At higher values of pressure, the experimental results showed a more remarkable similarity to the results from simulation (M1, M2, and M3), resulting from loss of slackness of the pneumatic

actuator with an increase in pressure, thus producing a stable and close to ideal results.

- 2 The plots reveal that the model M2 produced a trend that closely resembled the experimental results compared to the other two models (M1 and M3). Model M2 considers the bladder's hyper-elastic nature and the braided sleeve's contribution, thus accounting for a more realistic trend compared to M1 and M3.

Authors contributions

Antonio Dylan Do Rosario Carvalho: Conceptualization, Methodology, Formal analysis, Investigation, Experimentation, Writing - Original Draft, Writing - Review & Editing, Visualization.

Navin Karanth P: Conceptualization, Methodology, Formal analysis, Writing - Review & Editing Supervision.

Vijay Desai: Conceptualization, Methodology, Formal analysis, Writing - Review & Editing, Supervision.

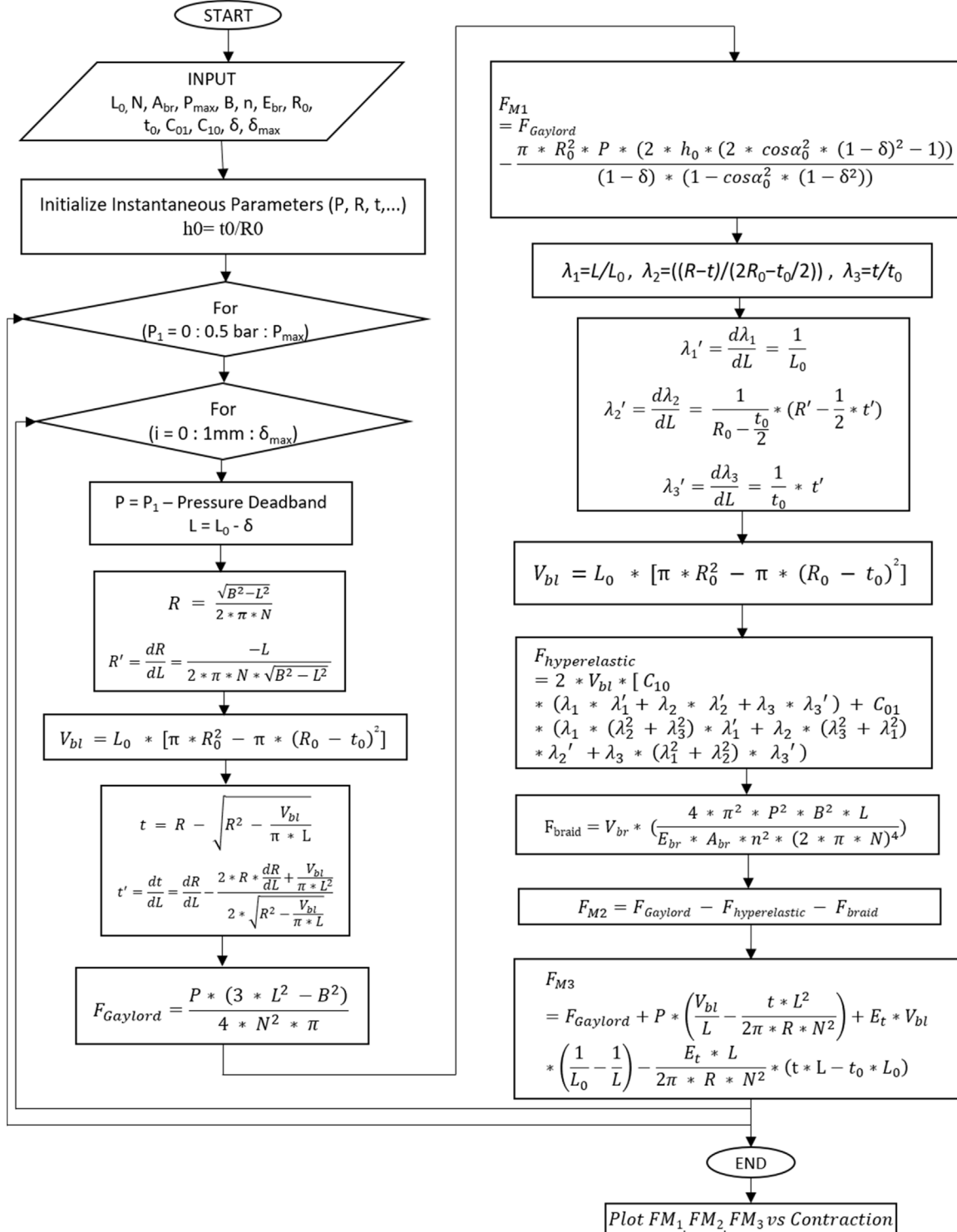
Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A.



Appendix B. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.sna.2021.112877>.

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